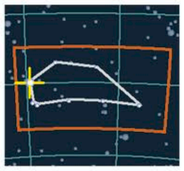
WHITE STAR

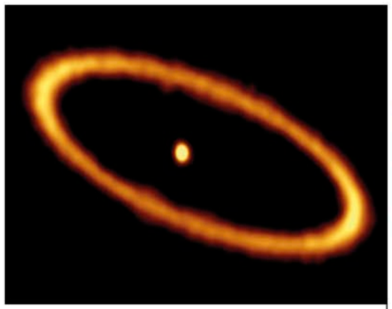
Fomalhaut



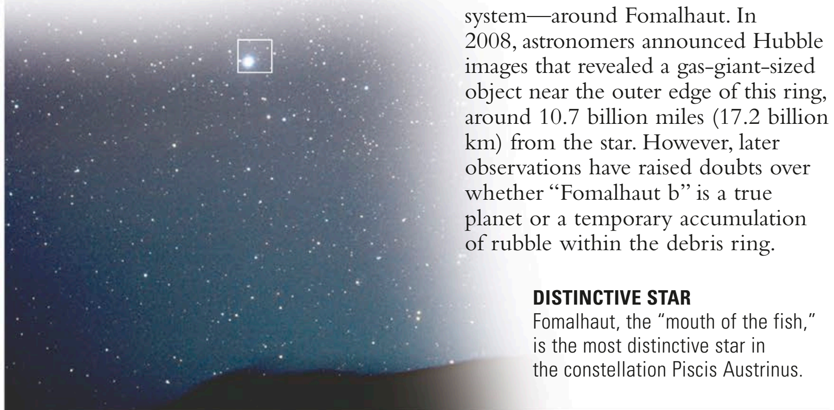
DISTANCE FROM SUN	25.1 light-years
MAGNITUDE	1.16
SPECTRAL TYPE	A

PISCIS AUSTRINUS

The brightest star in Piscis Austrinus, Fomalhaut is the 18th-brightest star in the sky. It has a surface temperature of about 15,000°F (8,500°C), with a luminosity 16 times that of the Sun. In 1983, the infrared telescope IRAS revealed that it was a source of greater infrared radiation than expected. Further observations revealed that the infrared radiation is being emitted by a ring of dust particles—with a diameter over twice that of the solar system—around Fomalhaut. In 2008, astronomers announced Hubble images that revealed a gas-giant-sized object near the outer edge of this ring, around 10.7 billion miles (17.2 billion km) from the star. However, later observations have raised doubts over whether "Fomalhaut b" is a true planet or a temporary accumulation of rubble within the debris ring.



GLOWING RING
This image from the ALMA radio telescope reveals Fomalhaut's sharply defined outer dust ring, which is thought to be kept in line by unseen planets.



DISTINCTIVE STAR
Fomalhaut, the "mouth of the fish," is the most distinctive star in the constellation Piscis Austrinus.

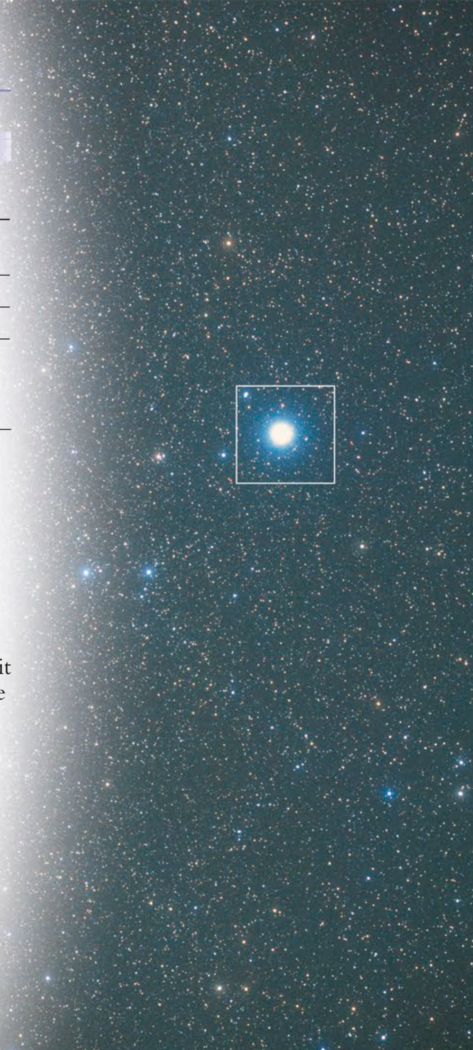
WHITE STAR

Vega

DISTANCE FROM SUN	25.3 light-years
MAGNITUDE	0.03
SPECTRAL TYPE	A

LYRA

Also known as Alpha (α) Lyrae, Vega is the fifth-brightest star in the sky. Along with Altair (opposite) and Deneb, it makes up the Summer Triangle. Vega has a mass of about 2.5 solar masses, a luminosity 54 times that of the Sun, and a surface temperature of about 16,500°F (9,300°C). Around 12,000 years ago, it was the north Pole Star, and it will be so again in about 14,000 years. In 1983, the infrared satellite IRAS revealed that it is surrounded by a disk of dusty material that is possibly the precursor to a planetary system. Vega is the ultimate "standard" star used to calibrate the spectral range and apparent magnitude of stars in optical astronomy (see p.233).



BRIGHT BEACON
The brightest star in the northern summer sky, Vega takes its name from an Arabic word meaning "swooping eagle."

YELLOW-WHITE STAR

Porrima

DISTANCE FROM SUN	38 light-years
MAGNITUDE	2.74
SPECTRAL TYPE	F

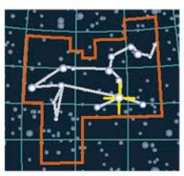
VIRGO

THE PORRIMA PAIR

Porrima, also known as Gamma (γ) Virginis, is a binary system made up of two almost identical stars, both about 1.5 times the mass of the Sun. Their surface temperatures are around 13,000°F (7,000°C), and they appear creamy white in amateur telescopes. Their luminosities are each about four times that of the Sun. They orbit each other in a very elliptical path that takes around 170 years to complete.

BLUE-WHITE STAR

Regulus

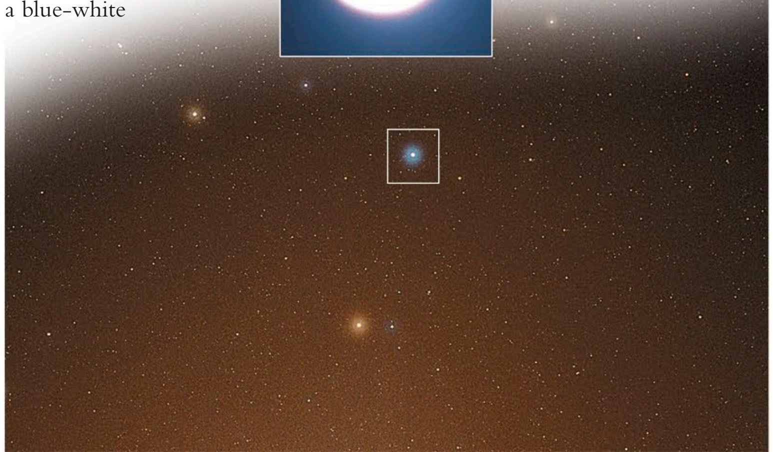


DISTANCE FROM SUN	78 light-years
MAGNITUDE	1.35
SPECTRAL TYPE	B

LEO

The brightest star in the constellation Leo, Regulus just makes it into the top 25 brightest stars as seen from Earth. *Regulus* is a Latin word meaning "little king." The star is situated at the base of the distinctive sickle asterism (shaped like a reversed question mark) in the constellation. It lies very close to the ecliptic (see pp.62–65) and is often occulted by the Moon (right). Regulus is a triple system. The brightest component is a blue-white main-sequence star about 3.5 times the mass of the Sun and with a diameter also around 3.5 times that of the Sun. It has a surface temperature of about 22,000°F (12,000°C) and shines at about 140 times the brightness of the Sun. It is also an emitter of high levels of ultraviolet radiation. Regulus has a companion binary star composed of an orange dwarf and a red dwarf separated by about 9 billion miles (14 billion km). These dwarf components orbit each other over a period of about 1,000 years, and they in turn orbit the main star once every 130,000 years.

REGULUS OCCULTED
Poised at the top-left curve of the Moon, Regulus is about to be occulted as the Moon passes in front of it. Occultations can help astronomers determine the diameters of large stars and ascertain whether they are binary systems. Occultations by the Moon can also reveal details about the Moon's surface features.



BLUE STAR

Gamma Velorum

DISTANCE FROM SUN	840 light-years
MAGNITUDE	1.8
SPECTRAL TYPES	O and WR

VELA

This blue star is also sometimes known as Regor—"Roger" spelled backward—in honor of the astronaut Roger Chaffee, who died in a fire during a routine test onboard the Apollo 1 spacecraft in 1967. Gamma (γ) Velorum is a complex star system dominated by a blue subgiant poised to evolve off the main sequence. Its evolution has been affected by being in a very close binary orbit with a star that is now a Wolf–Rayet star. They lie as close as Earth does to the Sun and orbit each other every 78.5 days. The Wolf–Rayet star is now the less massive component of the close binary, but probably started as the more massive and evolved much more rapidly. The subgiant has around 30 times the mass of the Sun, with a surface temperature of 60,000°F (35,000°C) and a luminosity around 200,000 times that of the Sun. There are also two other components to the system lying much farther away, one of which is a hot B-type star (see pp.232–233) at a distance of about 0.16 light-years.